

Assignment

Date _____ Period _____

Simplify. Write your answer in rectangular form.

1) $4\left(\cos \frac{2\pi}{3} + i\sin \frac{2\pi}{3}\right) \cdot 6\left(\cos \frac{4\pi}{3} + i\sin \frac{4\pi}{3}\right)$

2) $6\left(\cos \frac{\pi}{6} + i\sin \frac{\pi}{6}\right) \cdot 4\left(\cos \frac{5\pi}{4} + i\sin \frac{5\pi}{4}\right)$

Product theorem: multiply the r's, add the angles.

$$r = 4 \cdot 6 = 24$$

$$\theta = \frac{2\pi}{3} + \frac{4\pi}{3} = \frac{6\pi}{3} = 2\pi$$

 2π is a full turn, so it lands on 0. Keep angles in $[0, 2\pi)$.

$$24(\cos 0 + i\sin 0) = 24(1 + 0i)$$

$$\boxed{24}$$

*A product can land exactly on the real axis. $24 + 0i$ is just 24.**Multiply r's, add angles (common denominator 12).*

$$r = 6 \cdot 4 = 24 \quad \theta = \frac{2\pi}{12} + \frac{15\pi}{12} = \frac{17\pi}{12} \quad (255^\circ)$$

 255° is not on the unit circle. Split it into two angles that are: *$\frac{17\pi}{12} = \frac{7\pi}{6} + \frac{\pi}{4}$ ($210^\circ + 45^\circ$), then use the sum formulas.*

$$\cos \frac{17\pi}{12} = \cos \frac{7\pi}{6} \cos \frac{\pi}{4} - \sin \frac{7\pi}{6} \sin \frac{\pi}{4}$$

$$= \left(-\frac{\sqrt{3}}{2}\right) \frac{\sqrt{2}}{2} - \left(-\frac{1}{2}\right) \frac{\sqrt{2}}{2} = \frac{\sqrt{2} - \sqrt{6}}{4}$$

$$\sin \frac{17\pi}{12} = \sin \frac{7\pi}{6} \cos \frac{\pi}{4} + \cos \frac{7\pi}{6} \sin \frac{\pi}{4}$$

$$= \left(-\frac{1}{2}\right) \frac{\sqrt{2}}{2} + \left(-\frac{\sqrt{3}}{2}\right) \frac{\sqrt{2}}{2} = \frac{-\sqrt{2} - \sqrt{6}}{4}$$

$$24 \left(\frac{\sqrt{2} - \sqrt{6}}{4} + i \frac{-\sqrt{2} - \sqrt{6}}{4} \right)$$

$$\boxed{6\sqrt{2} - 6\sqrt{6} + (-6\sqrt{6} - 6\sqrt{2})i}$$

Check: 255° is QIII, so both parts should be negative ($6\sqrt{2} - 6\sqrt{6} \approx -6.2$). They are.

3) $(2 - i)(-2 + 5i)$

Product theorem: multiply the r's, add the angles. These angles aren't on the unit circle, so don't find them in degrees (that needs a calculator). Call them θ_1 and θ_2 and use their exact $\cos \theta = \frac{x}{r}$ and $\sin \theta = \frac{y}{r}$.

$$2 - i: r_1 = \sqrt{2^2 + (-1)^2} = \sqrt{5}, \quad \cos \theta_1 = \frac{2}{\sqrt{5}}, \quad \sin \theta_1 = \frac{-1}{\sqrt{5}} \quad (QIV)$$

$$-2 + 5i: r_2 = \sqrt{(-2)^2 + 5^2} = \sqrt{29}, \quad \cos \theta_2 = \frac{-2}{\sqrt{29}}, \quad \sin \theta_2 = \frac{5}{\sqrt{29}} \quad (QII)$$

Product theorem: $r = r_1 r_2 = \sqrt{5} \cdot \sqrt{29} = \sqrt{145}$, angle = $\theta_1 + \theta_2$ (sum formulas give its cos and sin)

$$\cos(\theta_1 + \theta_2) = \cos \theta_1 \cos \theta_2 - \sin \theta_1 \sin \theta_2 = \frac{2}{\sqrt{5}} \cdot \frac{-2}{\sqrt{29}} - \frac{-1}{\sqrt{5}} \cdot \frac{5}{\sqrt{29}} = \frac{-4+5}{\sqrt{145}} = \frac{1}{\sqrt{145}}$$

$$\sin(\theta_1 + \theta_2) = \sin \theta_1 \cos \theta_2 + \cos \theta_1 \sin \theta_2 = \frac{-1}{\sqrt{5}} \cdot \frac{-2}{\sqrt{29}} + \frac{2}{\sqrt{5}} \cdot \frac{5}{\sqrt{29}} = \frac{2+10}{\sqrt{145}} = \frac{12}{\sqrt{145}}$$

$$\sqrt{145} \left(\frac{1}{\sqrt{145}} + \frac{12}{\sqrt{145}} i \right)$$

$$\boxed{1 + 12i}$$

The $\sqrt{145}$ cancels, which is why the answer comes out clean. Check: FOIL gives the same $1 + 12i$.

$$4) \frac{-5 + 4i}{4 + 5i}$$

Quotient theorem: divide the r's, subtract the angles. Keep $\cos \theta = \frac{x}{r}$, $\sin \theta = \frac{y}{r}$ exact.

$$\text{Top: } r_1 = \sqrt{25 + 16} = \sqrt{41}$$

$$\cos \theta_1 = \frac{-5}{\sqrt{41}}, \sin \theta_1 = \frac{4}{\sqrt{41}} \text{ (QII)}$$

$$\text{Bottom: } r_2 = \sqrt{16 + 25} = \sqrt{41}$$

$$\cos \theta_2 = \frac{4}{\sqrt{41}}, \sin \theta_2 = \frac{5}{\sqrt{41}} \text{ (QI)}$$

$$r = \frac{\sqrt{41}}{\sqrt{41}} = 1, \text{ angle} = \theta_1 - \theta_2$$

$$\cos(\theta_1 - \theta_2) = \cos \theta_1 \cos \theta_2 + \sin \theta_1 \sin \theta_2$$

$$= \frac{-20}{41} + \frac{20}{41} = 0$$

$$\sin(\theta_1 - \theta_2) = \sin \theta_1 \cos \theta_2 - \cos \theta_1 \sin \theta_2$$

$$= \frac{16}{41} - \frac{-25}{41} = \frac{41}{41} = 1$$

$$1(0 + 1i)$$

$$\boxed{i}$$

$\cos = 0$, $\sin = 1$ means the angle difference is exactly 90° :
 $1 \text{ cis } 90^\circ = i$.

$$5) \frac{4 + 3i}{2i}$$

Quotient theorem. The bottom sits on the imaginary axis, so it is already easy: $2i = 2 \text{ cis } 90^\circ$.

$$\text{Top: } r_1 = \sqrt{16 + 9} = 5, \cos \theta_1 = \frac{4}{5}, \sin \theta_1 = \frac{3}{5}$$

$$\text{Bottom: } r_2 = 2, \theta_2 = 90^\circ$$

$$r = \frac{5}{2}, \text{ angle} = \theta_1 - 90^\circ$$

$$\cos(\theta_1 - 90^\circ) = \cos \theta_1 \cos 90^\circ + \sin \theta_1 \sin 90^\circ$$

$$= \frac{4}{5}(0) + \frac{3}{5}(1) = \frac{3}{5}$$

$$\sin(\theta_1 - 90^\circ) = \sin \theta_1 \cos 90^\circ - \cos \theta_1 \sin 90^\circ$$

$$= \frac{3}{5}(0) - \frac{4}{5}(1) = -\frac{4}{5}$$

$$\frac{5}{2} \left(\frac{3}{5} - \frac{4}{5}i \right) = \frac{15}{10} - \frac{20}{10}i$$

$$\boxed{\frac{3}{2} - 2i}$$

Subtracting 90° swaps cos and sin and flips a sign. Distribute $\frac{5}{2}$ to both parts.

$$6) \frac{4 - 6i}{-6 - 2i}$$

Quotient theorem with exact cos and sin.

$$\text{Top: } r_1 = \sqrt{16 + 36} = \sqrt{52} = 2\sqrt{13}$$

$$\cos \theta_1 = \frac{4}{2\sqrt{13}} = \frac{2}{\sqrt{13}}, \sin \theta_1 = \frac{-3}{\sqrt{13}} \text{ (QIV)}$$

$$\text{Bottom: } r_2 = \sqrt{36 + 4} = \sqrt{40} = 2\sqrt{10}$$

$$\cos \theta_2 = \frac{-3}{\sqrt{10}}, \sin \theta_2 = \frac{-1}{\sqrt{10}} \text{ (QIII)}$$

$$r = \frac{2\sqrt{13}}{2\sqrt{10}} = \frac{\sqrt{13}}{\sqrt{10}}$$

$$\cos(\theta_1 - \theta_2) = \frac{2}{\sqrt{13}} \cdot \frac{-3}{\sqrt{10}} + \frac{-3}{\sqrt{13}} \cdot \frac{-1}{\sqrt{10}} = \frac{-3}{\sqrt{130}}$$

$$\sin(\theta_1 - \theta_2) = \frac{-3}{\sqrt{13}} \cdot \frac{-3}{\sqrt{10}} - \frac{2}{\sqrt{13}} \cdot \frac{-1}{\sqrt{10}} = \frac{11}{\sqrt{130}}$$

$$\frac{\sqrt{13}}{\sqrt{10}} \cdot \frac{-3}{\sqrt{130}} = \frac{-3\sqrt{13}}{\sqrt{1300}} = \frac{-3\sqrt{13}}{10\sqrt{13}} = -\frac{3}{10}$$

$$\boxed{-\frac{3}{10} + \frac{11}{10}i}$$

Same cancel for the imaginary part: $\frac{11\sqrt{13}}{10\sqrt{13}} = \frac{11}{10}$.

$$7) (6(\cos 30 + i \sin 30))^4$$

De Moivre: raise r to the 4th, multiply the angle by 4. The angle is in degrees, so the answer stays in degrees.

$$6^4 = 1296 \quad 4 \cdot 30^\circ = 120^\circ$$

$$1296(\cos 120^\circ + i \sin 120^\circ)$$

$$= 1296 \left(-\frac{1}{2} + \frac{\sqrt{3}}{2}i \right)$$

$$\boxed{-648 + 648\sqrt{3}i}$$

120° is QII: cos negative, sin positive. Distribute 1296 to both parts.

$$8) (5 - 3i)^4$$

De Moivre: $r^4 \text{cis} 4\theta$. θ isn't on the unit circle, so get $\cos 4\theta$ and $\sin 4\theta$ with the double-angle formulas, twice.

$$r = \sqrt{25 + 9} = \sqrt{34}, \quad r^4 = 34^2 = 1156$$

$$\cos \theta = \frac{5}{\sqrt{34}}, \quad \sin \theta = \frac{-3}{\sqrt{34}} \quad (\text{QIV})$$

$$\cos 2\theta = \cos^2 \theta - \sin^2 \theta = \frac{25-9}{34} = \frac{8}{17}$$

$$\sin 2\theta = 2 \sin \theta \cos \theta = \frac{2(-3)(5)}{34} = -\frac{15}{17}$$

$$\cos 4\theta = \cos^2 2\theta - \sin^2 2\theta = \frac{64-225}{289} = -\frac{161}{289}$$

$$\sin 4\theta = 2 \sin 2\theta \cos 2\theta = \frac{2(-15)(8)}{289} = -\frac{240}{289}$$

$$1156 \left(-\frac{161}{289} - \frac{240}{289}i \right) \quad (1156 = 4 \cdot 289)$$

$$\boxed{-644 - 960i}$$

$$9) (5 - 6i)^4$$

Same De Moivre move: $r^4 \text{cis} 4\theta$, double-angle formulas twice.

$$r = \sqrt{25 + 36} = \sqrt{61}, \quad r^4 = 61^2 = 3721$$

$$\cos \theta = \frac{5}{\sqrt{61}}, \quad \sin \theta = \frac{-6}{\sqrt{61}} \quad (\text{QIV})$$

$$\cos 2\theta = \frac{25-36}{61} = -\frac{11}{61}$$

$$\sin 2\theta = \frac{2(-6)(5)}{61} = -\frac{60}{61}$$

$$\cos 4\theta = \frac{(-11)^2 - (-60)^2}{3721} = \frac{121-3600}{3721} = -\frac{3479}{3721}$$

$$\sin 4\theta = \frac{2(-60)(-11)}{3721} = \frac{1320}{3721}$$

$$3721 \left(-\frac{3479}{3721} + \frac{1320}{3721}i \right)$$

$$\boxed{-3479 + 1320i}$$

r^4 always cancels the denominator. $2(-60)(-11)$ is positive: two negatives.

Find all n th roots. Write your answers in polar form.

$$10) 1 + i\sqrt{3}, \quad n = 5$$

Convert to trig form, then the n th root theorem:

$$\sqrt[n]{r} \text{cis} \left(\frac{\theta}{n} + \frac{2\pi k}{n} \right).$$

$$r = \sqrt{1^2 + (\sqrt{3})^2} = \sqrt{4} = 2$$

$$\tan \theta = \frac{\sqrt{3}}{1} = \sqrt{3}, \quad \text{QI, so } \theta = \frac{\pi}{3}$$

$$\sqrt[5]{2} \quad \text{first: } \frac{\pi/3}{5} = \frac{\pi}{15} \quad \text{spacing: } \frac{2\pi}{5} = \frac{6\pi}{15}$$

Common denominator 15, then keep adding $\frac{6\pi}{15}$ until you have 5 roots.

$$\sqrt[5]{2} \text{cis} \frac{\pi}{15}, \sqrt[5]{2} \text{cis} \frac{7\pi}{15}, \sqrt[5]{2} \text{cis} \frac{13\pi}{15},$$

$$\sqrt[5]{2} \text{cis} \frac{19\pi}{15}, \sqrt[5]{2} \text{cis} \frac{5\pi}{3}$$

$\frac{25\pi}{15} = \frac{5\pi}{3}$. In degrees: $12^\circ, 84^\circ, 156^\circ, 228^\circ, 300^\circ$ (same roots).

$$11) -\frac{5}{2} + \frac{5\sqrt{3}}{2}i, \quad n = 5$$

Convert first. Both parts have a denominator of 2, so square carefully.

$$r = \sqrt{\left(-\frac{5}{2}\right)^2 + \left(\frac{5\sqrt{3}}{2}\right)^2} = \sqrt{\frac{25}{4} + \frac{75}{4}} = \sqrt{25} = 5$$

$$\tan \theta = \frac{5\sqrt{3}/2}{-5/2} = -\sqrt{3}, \quad \text{ref } 60^\circ, \text{ QII, so } \theta = 120^\circ$$

$$\sqrt[5]{5} \quad \text{first: } \frac{120^\circ}{5} = 24^\circ \quad \text{spacing: } \frac{360^\circ}{5} = 72^\circ$$

$$\sqrt[5]{5} \text{cis} 24^\circ, \sqrt[5]{5} \text{cis} 96^\circ, \sqrt[5]{5} \text{cis} 168^\circ,$$

$$\sqrt[5]{5} \text{cis} 240^\circ, \sqrt[5]{5} \text{cis} 312^\circ$$

$$\left(\frac{5\sqrt{3}}{2}\right)^2 = \frac{25 \cdot 3}{4} = \frac{75}{4}. \quad \sqrt[5]{5} \text{ doesn't simplify, so leave it.}$$

$$12) \frac{5}{2} - \frac{5\sqrt{3}}{2}i, n = 2$$

Square roots ($n = 2$): two answers, 180° apart.

$$r = \sqrt{\left(\frac{5}{2}\right)^2 + \left(-\frac{5\sqrt{3}}{2}\right)^2} = \sqrt{\frac{25}{4} + \frac{75}{4}} = \sqrt{25} = 5$$

$$\tan \theta = \frac{-5\sqrt{3}/2}{5/2} = -\sqrt{3}, \text{ ref } 60^\circ, \text{ QIV, so } \theta = 300^\circ \quad (\text{real part positive, imaginary part negative})$$

$$\sqrt{5} \quad \text{first: } \frac{300^\circ}{2} = 150^\circ \quad \text{spacing: } \frac{360^\circ}{2} = 180^\circ, \text{ so } 150^\circ + 180^\circ = 330^\circ$$

$$\boxed{\sqrt{5} \text{cis } 150^\circ, \quad \sqrt{5} \text{cis } 330^\circ}$$

Take the root of r ($\sqrt{5}$), not half of it. The two square roots always point in opposite directions.

Find all roots. Answer in standard form.

$$13) x^3 - 64 = 0$$

Isolate x^3 , write the constant in trig form, take cube roots, then convert each root to standard form.

$$x^3 = 64 = 64 + 0i: \text{ positive real axis, so } r = 64, \theta = 0^\circ$$

$$\sqrt[3]{64} = 4 \quad \text{first: } \frac{0^\circ}{3} = 0^\circ \quad \text{spacing: } \frac{360^\circ}{3} = 120^\circ \quad \text{roots: } 4 \text{cis } 0^\circ, 4 \text{cis } 120^\circ, 4 \text{cis } 240^\circ$$

$$4 \text{cis } 0^\circ = 4 \quad 4 \text{cis } 120^\circ = 4 \left(-\frac{1}{2} + \frac{\sqrt{3}}{2}i \right) = -2 + 2i\sqrt{3} \quad 4 \text{cis } 240^\circ = 4 \left(-\frac{1}{2} - \frac{\sqrt{3}}{2}i \right) = -2 - 2i\sqrt{3}$$

$$\boxed{\left\{ 4, -2 + 2i\sqrt{3}, -2 - 2i\sqrt{3} \right\}}$$

The n th root theorem gives exactly 3 cube roots, so $x = 4$ alone is only a third of the answer.

Answers to Assignment (ID: 1)

- | | | |
|--|---|--|
| 1) 24 | 2) $6\sqrt{2} - 6\sqrt{6} + (-6\sqrt{6} - 6\sqrt{2}) \cdot i$ | 3) $1 + 12i$ |
| 4) i | 5) $\frac{3}{2} - 2i$ | 6) $-\frac{3}{10} + \frac{11}{10}i$ |
| 8) $-644 - 960i$ | 9) $-3479 + 1320i$ | 10) $\sqrt[5]{2} \left(\cos \frac{\pi}{15} + i \sin \frac{\pi}{15} \right)$
$\sqrt[5]{2} \left(\cos \frac{7\pi}{15} + i \sin \frac{7\pi}{15} \right)$
$\sqrt[5]{2} \left(\cos \frac{13\pi}{15} + i \sin \frac{13\pi}{15} \right)$
$\sqrt[5]{2} \left(\cos \frac{19\pi}{15} + i \sin \frac{19\pi}{15} \right)$
$\sqrt[5]{2} \left(\cos \frac{5\pi}{3} + i \sin \frac{5\pi}{3} \right)$ |
| | | 11) $\sqrt[5]{5}(\cos 24 + i \sin 24)$
$\sqrt[5]{5}(\cos 96 + i \sin 96)$
$\sqrt[5]{5}(\cos 168 + i \sin 168)$
$\sqrt[5]{5}(\cos 240 + i \sin 240)$
$\sqrt[5]{5}(\cos 312 + i \sin 312)$ |
| 12) $\sqrt{5}(\cos 150 + i \sin 150)$
$\sqrt{5}(\cos 330 + i \sin 330)$ | 13) $\{4, -2 + 2i\sqrt{3}, -2 - 2i\sqrt{3}\}$ | |

Every worked answer on pages 1–4 matches this key. #10 is given in radians here; the degree version ($12^\circ, 84^\circ, 156^\circ, 228^\circ, 300^\circ$) is the same set of roots.